MOHAMMAD ARSHAD RAHMAN

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APPOINTMENTS

Associate Professor, Economic Sciences, IIT Kanpur, India (Dec, 2019 – present)

Previous Positions

Associate Professor, College of Business, Zayed University, UAE (Dec, 2020 – Dec, 2022)
Assistant Professor, Economic Sciences, IIT Kanpur, India (May, 2017 – Dec, 2019)
Assistant Professor of Economics, HSS, IIT Kanpur, India (Jul, 2013 – May, 2017)
Business Analyst, Hewlett-Packard DSAS, India (Feb, 2007 – Aug, 2008)
Economic Analyst, Roulac Global Places LLC, India (Jun, 2006 – Jan, 2007)
Summer Intern, National Council of Applied Economic Research, India (Apr, 2005 – Jun, 2005)

Visiting Positions

Visiting Scientist, EPU, Indian Statistical Institute Delhi, India (Oct, 2023 – Nov, 2023)
Lecturer, Department of Economics, University of California, Irvine, USA (Jun, 2014 – Jul, 2014)

EDUCATION

Ph.D., Economics, University of California, Irvine, USA (2008 – 2013)
Dissertation Title: *Explorations in Simulation Based Techniques*
Advisors: Ivan Jeliazkov (Chair), Dale J. Poirier, David Brownstone, Fabio Milani
Master of Science, Statistics, University of California, Irvine, USA (2009 – 2011)
Master of Arts, Economics, University of California, Irvine, USA (2008 – 2009)
Master of Arts, Economics, Delhi School of Economics, University of Delhi, India (2004 – 2006)
Bachelor of Science, Economics, St. Xavier's College, University of Calcutta, India (2001 – 2004)
W.B.H.S.E, Class XII, St. Xavier's College, University of Calcutta, India (1999 – 2001)

RESEARCH

Theoretical	Bayesian econometrics, quantile regression, discrete choice models, Markov chain Monte Carlo (MCMC) methods, machine learning, and shrinkage & variable selection.
Applied	Applications in the social sciences, agriculture, energy and environmental economics, finance, housing, labor economics, political science, transportation, and engineering.

TEACHING

Indian Institute of Technology Kanpur (IIT Kanpur), India

Graduate	Econometrics (2), Bayesian Econometrics (1).
Undergraduate	Bayesian Data Analysis (4), Bayesian Econometrics (7), Bayesian Methods in Finance (1), Econometrics I (12), Econometrics II (4), International Economics and Finance (1), Panel Data Procedures and Analysis (1).

e-Masters Data Analysis in R and Python (6).

Zayed University, Abu Dhabi, United Arab Emirates

Undergraduate Financial Markets (5), Financial Planning and Forecasting (2), Introduction to Finance (7).

Chennai Mathematical Institute, Chennai, India

Masters Econometrics (2).

University of California, Irvine, California, United States

Undergraduate Probability & Statistics (1).

Note: Number within parenthesis indicates the number of times I have taught the course.

ADVISING

Ph.D. Students Prashant Jha (2017− 2019), Soumyajit Bhunia (2026−present).

M.S. Students Navya Nihal (2026), Abhiraj Akhouri (2025−2026), Lalit Upadhyay (2025−2026), Arjun Gupta (2019−2020), Mayank Garg (2018−2019), Amritesh Kumar (2018−2019), Om Prakash Bankolia (2018−2019), Prasoon Bajpai (2018−2019), Rahul Sharma (2017−2018), Shubham Karnawat (2016−2017), Sanket Jantre (2016−2017), Abhishek Srivastava (2016−2017), Mohit Batham (2015−2016), Pranay Jain (2015−2016), Alisha Sarkar (2015−2016).

Undergraduate Students Yeva Gupta (2026), Rhut Patel (2025), Shrish Shete (2024), Mohd Siraj (SURGE, 2023), Pulkit Dhamija (2023), Ritika Ajmera (2020), Apoorva Sheera (2019−2020), Arjun Gupta (2018−2019), Anshul Goel (2018), Aayushi Pandey (2018), Debasmita Das (SURGE, 2017), Sanjana Shukla (2017), Rahul Sharma (2016−2017), Shrenik Bohra (2015−2016), Isha Gupta (SURGE, 2015), Mohit Batham (2014−2015), Nafisa Lohawala (2014−2015).

PUBLICATIONS

(1) "Bayesian Analysis of Multivariate Sample Selection Models Using Gaussian Copulas," with Phillip Li, *Advances in Econometrics*, 27, 269−288, 2011. (Journal ranking: Scimago−Q2, 2011)

(2) "Binary and Ordinal Data Analysis in Economics: Modeling and Estimation," with Ivan Jeliazkov, *Mathematical Modeling with Multidisciplinary Applications*, edited by X.S. Yang, 123−150, 2013. John Wiley & Sons Inc, Hoboken, New Jersey.

(3) "Quantile Regression using Metaheuristic Algorithms," *International Journal of Computational Economics and Econometrics*, 3(3/4), 205−233, 2013. (Journal rankings: ABDC−C, Scimago−Q4)

(4) "Bayesian Quantile Regression for Ordinal Models," *Bayesian Analysis*, 11(1), 1−24, 2016. (Journal rankings: ABDC−A, Scimago−Q1)

Associated R-package: bqror.

(5) "To Drill or Not to Drill? An Econometric Analysis of US Public Opinion," with Deep Mukherjee, *Energy Policy*, 91, 341−351, 2016. (Journal rankings: ABDC−A, Scimago−Q1)

Media Coverage: Journalist's Resource

(6) "Are Strategies for Success Different in Test Cricket and One-day Internationals? Evidence from England-Australia Rivalry," with Nafisa Lohawala†, *Journal of Sports Analytics*, 4(3), 175−191, 2018.

(7) "Estimation and Applications of Quantile Regression for Binary Longitudinal Data," with Angela Vossmeyer, *Advances in Econometrics*, 40B, 157−191, 2019. (Journal ranking: Scimago−Q1, 2019)

Associated R-package: qbld.

(8) "Flexible Bayesian Quantile Regression in Ordinal Models," with Shubham Karnawat†, *Advances in Econometrics*, 40B, 211−251, 2019. (Journal ranking: Scimago−Q1, 2019)

(9) "Bayesian Updating of Structural Model with a Conditionally Heteroscedastic Error Distribution," with Gideon Arthur Lyngdoh† and Sudib Kumar Mishra, *Journal of Engineering Mechanics–ASCE*, 145(12), 1-18, 2019 (Article No. 04019091). (Journal ranking: Scimago−Q1)

(10) "Bayesian Panel Quantile Regression for Binary Outcomes with Correlated Random Effects: An Application on Crime Recidivism in Canada," with Georges Bresson and Guy Lacroix, *Empirical Economics*, 60(1), 127-159, 2021. (Journal rankings: ABDC−A, Scimago−Q1)

(The article was published in a special issue honoring Professor Badi Baltagi's contribution to the field of econometrics.)

(11) "Seemingly Unrelated Regression with Measurement Error: Estimation via Markov chain Monte Carlo and Mean Field Variational Bayes Approximation," with Georges Bresson, Anoop Chaturvedi, and Shalabh, *International Journal of Biostatistics*, 17(1), 75-97, 2021. (Journal rankings: Scimago−Q2)

(12) "Do Online Courses Provide an Equal Educational Value Compared to In-Person Classroom Teaching? Evidence from U.S. Survey Data using Quantile Regression," with Manini Ojha, *Education Policy Analysis Archives*, 29(85), 1-25, 2021. (Journal ranking: Scimago−Q2)

(13) "Heterogeneity in Food Expenditure amongst US families: Evidence from Longitudinal Quantile Regression," with Arjun Gupta† and Soudeh Mirghasemi, *Indian Economic Review*, 56(1), 25-48, 2021. (Journal ranking: ABDC−C, Scimago−Q1)

(14) "bqror: An R package for Bayesian Quantile Regression in Ordinal Models," with Prajual Maheshwari†, *The R Journal*, 15(2), 39-55, 2023. (Journal ranking: Scimago−Q1)

(15) "Binary and Ordinal Probit Regression: Applications to Public Opinion on Marijuana Legalization in the United States," with Mohit Batham†, Manini Ojha and Soudeh Mirghasemi, *Applied Econometric Analysis using Cross Section and Panel Data*, edited by Deep Mukherjee, 33-60, 2023. Contribution to Economics, Springer Singapore, Singapore.

(16) "Do Determinants of EV Purchase Intent Vary Across the Spectrum? Evidence from Bayesian Analysis of US Survey Data," with Nafisa Lohawala, *Transportation Research Part A: Policy and Practice*, 207, 1-19, 2026 (Article No. 104961). (Journal rankings: ABDC−A*, Scimago−Q1)

This paper was earlier circulated with the title, To Buy an Electric Vehicle or Not? A Bayesian Analysis of Consumer Intent in the United States.

(17) "A New Media-Based Framework for Climate Risk: Evidence from Television Coverage and Clean Energy Firms," with Suruchi Shrimali, Preeti Roy, Wasim Ahmad, and Vipin, *International Review of Financial Analysis*, 120, 1-19, 2026 (Article No. 105500). (Journal rankings: ABDC−A, Scimago−Q1).

The paper was earlier circulated with the title, "Tuning into Climate Risks: Extracting Innovation from Television News for Clean Energy Firms".

† denotes that the co-author is/was my student at the time of writing the paper.

WORKING PAPERS

• "Flexible Quantile Regression for Panel Data: Modeling, Estimation, and Application to US Rental Markets," with Ivan Jeliazkov, Shubham Karnawat†, and Angela Vossmeyer (*R & R, Bayesian Analysis*).

• "Modeling Misclassification in Spousal Violence Reporting: Evidence from Bayesian Quantile Regression," with Joon Jin Song, Yoo-Mi Chin, and James D. Stamey (*Under Review*).

• "Spatial Spillovers in Emergency Lending during the Great Depression: Evidence from Censored Panel Quantile Regression," with Oncel Aldanmaz and Angela Vossmeyer.

WORK IN PROGRESS

• "Challenges to Viksit Bharat: Evaluating the Downside Risks to India's Economic Growth", with Abhiraj Akhouri.

• "Bayesian Quantile Analysis of Semiparametric Additive Models for Sequential Data," with Ivan Jeliazkov.

• "Bayesian Variable Selection: Different Methods and Applications".

• "What is the Impact of Zinc Use on Paddy Production? Evidence from Quantile Analysis using data from Andhra Pradesh, India," with Deep Mukherjee.

SEMINARS & CONFERENCE PRESENTATIONS

2026 IIM Calcutta Economics Research Conference, Indian Institute of Management, Calcutta (Kolkata), India (Mar); Third Amitava Bose Memorial Annual Workshop, Indian Institute of Management, Calcutta (Kolkata), India (Aug); Joint Research Meeting, ISBIS StatFin 2026, Chennai Mathematical Institute, Chennai, India (Dec, scheduled).

2025 Fifth International Conference on Issues in Economic Theory and Policy, Presidency University, Kolkata, India (Dec).

2024 Asia Meeting of the Econometric Society (AMES-CSW 2024), Indian Institute of Technology Delhi, India (Jan); Centre for Mathematical and Computational Economics, Indian Institute of Technology Jodhpur, India (Apr); Eastern Asia Chapter of The International Society for Bayesian Analysis (EAC-ISBA), The Education University of Hong Kong, Hong Kong (Jun); 18th International Joint Conference CFE-CMStatistics, Kings College London, United Kingdom (Dec); Third Biennial Conference on Development: Energy, Environment & Climate Change, Indira Gandhi Institute of Development Research (IGIDR), Mumbai, India (Dec).

2023 Indian Institute of Technology Delhi, India (Aug); Department of Economics, Delhi School of Economics, India (Aug); Chennai Mathematical Institute, India (Sep); Economics and Planning Unit, Indian Statistical Institute Delhi, India (Nov).

2022 Statistical Methods in Finance, Chennai Mathematical Institute, India (Jun-Jul).

2021 Finance Research Seminar Series, College of Business, Zayed University, Abu Dhabi, UAE (Oct).

2020 International Virtual Conference on Statistics, University of Al-Qadisiyah, Iraq (May).

2019 Department of Economics, Ashoka University, India (Apr); Society for Economic Research in India (SERI) conference at the Indian Institute of Technology Kanpur, India (Jul); Winter School, Center for Development Economics, Delhi School of Economics, India (Dec), International Conference on Operations Research and Decision Sciences (ICORDS), Indian Institute of Management Visakhapatnam, India (Dec).

2018 Advances in Econometrics Conference, University of California, Irvine, USA (Jun); ASA Joint Statistical Meeting, Vancouver, Canada (Jul-Aug).

2017 Applied Statistics Division, Indian Statistical Institute Kolkata, India (Feb); Asian Meeting of the Econometric Society, Chinese University of Hong Kong, Hong Kong (Jun); Winter School, Center for Development Economics, Delhi School of Economics, India (Dec).

2016 Mini-symposium on Computation and Optimization, Department of Computer Science and Engineering, IIT Kanpur, India (Feb); Winter School, Center for Development Economics, Delhi School of Economics, India (Dec).

2015 Econometrics Seminar, Indian Institute of Management Indore, India (Apr); Econometrics Seminar, Department of Economics, University of California, Irvine, USA (Dec).

2013 Winter School, Center for Development Economics, Delhi School of Economics, India (Dec); Econometrics Seminar, Department of Economics, University of California, Irvine, USA (May).

2012 Econometrics Seminar, Department of Economics, University of California, Irvine, USA (Oct).

WORKSHOPS ORGANIZED

• "Statistical Techniques and R Software," (STARS) co-organized the workshop in collaboration with Shalabh, under the Quality Improvement Program (QIP) aimed at enhancing the skills and knowledge of faculty from AICTE recognized engineering institutions, IIT Kanpur, India.

(Workshop Dates: 8-12 Sep, 2016; 4-8 Dec, 2017; 3-7 Sep, 2018; 10-14 Feb, 2020)

• "Asset Pricing Meets Econometrics: Classical and Bayesian Regression in Practice," organized as a pre-conference workshop at the Joint Research Meeting, ISBIS StatFin 2026, Chennai Mathematical Institute, India, December 2026.

AWARDS, GRANTS & FELLOWSHIPS

Teaching Awards

Letter(s) of Appreciation for Excellence in Teaching from the Director, IIT Kanpur:

Bayesian Methods in Finance, ECO528 (Elective course)	(Sem II, 2025-26)
Econometrics II, ECO351 (Core course)	(Sem I, 2025-26)
Econometrics I, ECO251 (Core course)	(Summer, 2025)
Bayesian Econometrics, ECO545A (Elective course)	(Sem II, 2023-24)
Econometrics I, ECO341A (Core course)	(Summer, 2023)

Research Awards, Grants and Fellowships

Start-Up Grant Award, Zayed University, UAE	(2021 – 2022)
MATRICS, Science and Engineering Board, India (*Declined*, moved to UAE)	(2020)
Institute Fellowship, IIT Kanpur, India	(2013 – 2016)
Social Science Merit Fellowship, University of California (UC), Irvine, USA	(2008 – 2013)
Summer Research Fellowship, Department of Economics, UC Irvine, USA	(2009 – 2012)
Summer Stipend/Research Travel Grant, Dept. of Economics, UC Irvine, USA	(2009, 2010)
All India Post-Graduate Scholarship, University of Delhi, India	(2004 – 2006)

Corporate Awards

Analyst Accolade Award for work on Cluster Analysis, Hewlett Packard	(Apr, 2008)
e-Awards for annual contribution to USCB, Hewlett Packard	(2007)
e-Awards for contribution to consumer GTMU engagement, Hewlett Packard	(Nov, 2007)
Analyst Accolade Award, Sales-lift prediction analysis, Hewlett Packard	(Aug, 2007)

INSTITUTE/UNIVERSITY SERVICE

Indian Institute of Technology Kanpur (IIT Kanpur), India

Institute Service	2023–2024: Senate Undergraduate Committee (SUGC) Member; SUGC nominee to Senate Curriculum Development and Monitoring Committee (SCDMC).
	2019–2024: Center for Continuing Education (CCE) Advisory Board Member.
	2018–2023: Institute International Relations Committee (IIRC) Representative.
	2018–2019: Senate Undergraduate Committee (SUGC) Member.
	2014–2016: Global Strategy Ranking Committee Member.
Departmental Service	2026–2027: Economics Computer Lab In-charge; Department Finance Committee (Member).
	2025–2026: Economics Computer Lab In-charge; Student Recruitment Committee (Member).
	2024–2025: Departmental Undergraduate Committee (Member); Economics Computer Lab In-charge; Student Recruitment Committee (Member).
	2023–2024: Departmental Undergraduate Committee (Convener); Student Recruitment Committee (Member).
	2022–2023: Departmental Undergraduate Committee (Member).
	2020–2021: Student Placement Office (Departmental In-charge); Department Finance Committee (Member); Department Library Committee (Member).
	2019–2020: Departmental Undergraduate Committee (Member); Department Finance Committee (Convener); Web & Promotional Committee (Convener).
	2018–2019: Departmental Undergraduate Committee (Convener); Web & Promotional Committee (Convener).
	2017–2018: Space Committee (Convener); Web & Promotional Committee (Convener).
	2016–2017: Computer and Web Committee (In-charge); Department Faculty Recruitment Committee (Member).
	2015–2016: Economics Computer Lab In-charge.
	2014–2015: Research & Seminar Committee (Convener).
	2013–2014: Space Committee (Member).
JEE Duty	Institute Representative: Bhopal (2014, 2016); Gwalior (2015); Indore (2017, 2018, 2023); Kanpur (2019); Lucknow (2024, 2026).
GATE Duty	City Coordinator and Center Head: Aligarh (2016, 2026); Center Head: Aligarh (2015); Institute Representative: Aligarh (2017, 2018, 2025); Bareilly (2024).

Zayed University, Abu Dhabi, United Arab Emirates

Departmental Service	2022–2023: Teaching and Learning Committee (Member); Faculty Affairs Committee (Member).
	2021–2022: Teaching and Learning Committee (Member); Faculty Affairs Committee (Member).
	2020–2021: Assessment of Learning Committee (Member).

SERVICE TO PROFESSION

Referee Service for Books

Introduction to Financial Management, Cengage Learning (EMEA).

Referee Service for Journals

Advances in Data Analysis and Classification (1), Advances in Econometrics (3), Advances in Statistical Analysis (1), Agribusiness (2), Agricultural Economics (1), Annals of Applied Statistics (2), Applied Economics Letters (2), Bayesian Analysis (3), Behavior Research Methods (1), Biometrical Journal (1), British Food Journal (1), Canadian Journal of Statistics (1), Communication in Statistics−Theory and Methods (4), Communication in Statistics−Simulation and Computation (17), Computational Economics (6), Computational Statistics (5), Computational Statistics and Data Analysis (2), Data Science in Science (1), Econometric Reviews (1), Economics of Innovation and New Technology (1), Education Policy Analysis Archives (1), Engineering Reports (1), Environmental Impact Assessment Review (1), Frontiers in Public Health (1), IEEE Intelligent Systems (1), IEEE Transactions on Engineering Management (1), IEEE Transactions on Knowledge and Data Engineering (1), Indian Economic Review (1), International Journal of Mathematical Modelling and Numerical Optimization (1), International Journal of Public Opinion Research (1), International Review of Financial Analysis (1), Journal of Agricultural and Resource Economics (1), Journal of Applied Economics (1), Journal of Applied Statistics (5), Journal of Asian Business and Economic Studies (1), Journal of Computational and Applied Mathematics (1), Journal of Drug Issues (1), Journal of Econometrics (2), Journal of Economic Asymmetries (1), Journal of Probability and Statistics (1), Journal of Quantitative Economics (3), Journal of Research on Educational Effectiveness (1), Journal of Sports Sciences (1), Journal of Statistical Computation and Simulation (5), Journal of Statistical Software (1), Journal of Statistical Theory and Practice (1), Macroeconomics and Finance in Emerging Market Economies (1), Population Ecology (1), Sankhya B (3), Scandinavian Journal of Economics (1), Spatial Economic Analysis (1), Statistical Methods in Medical Research (1), Statistical Modelling (1), Statistical Papers (1), The Australian Journal of Agricultural and Resource Economics (1), Transportation Research Interdisciplinary Perspectives (1), US National Science Foundation (1), World Food Policy (1).

American Statistical Association (ASA) & Statistics Education

CAUSE Undergraduate Statistics Project Competition (USPROC) Judge on seven occasions.
(Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025, Spring 2026)

Association for Mentoring & Inclusion in Economics (AMIE)

Mentor, AMIE (2023-24)

SOFTWARE SKILLS

Statistical	Advance Knowledge of Matlab, R and SAS (Base, Stat, ETS, IML, GRAPH).
Office	WINDOWS, MS-OFFICE and Visual Basics. Received "Certificate of Excellence" from St. Xavier's College Computer Center (SXCCC).
Mark-up	LaTeX & Beamer (document preparation system for the TeX typesetting program).

AFFILIATIONS

American Economic Association, American Statistical Association, International Indian Statistical Association (IISA, Life Member), International Society for Bayesian Analysis (ISBA), International Society for Business and Industrial Statistics (ISBIS), International Statistical Institute (ISI), The Econometric

Society, The Indian Econometric Society (TIES, Life Member).

REFERENCES

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Claremont McKenna College
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